Geophysics III

Mixing Tutorial

1 Introduction

Most geologic materials are mixtures of either multiple minerals and/or fluids. However, when we make geological observations and model the fields we find that it is not possible to resolve these individual components. Instead, we are sensitive to the 'average' properties creating or modifying geophysical fields. Average is a bit of a dangerous word because it can indicate different things in different situations. So when someone says average I want you to ask "What kind?".

The typical person refers to the arithmetic mean when they use the term average. However, there are multiple types of averages that are mathematically defined. There is a mean, median and mode that you may remember from statistics. There are also an aritmetric, geometric, and harmonic means. Means can be weighted or unweighted. This tutorial focuses on each of these averages and when they may be appropriately applied to Earth's materials.

2 Statistical average

There are three averages that are typically used in statistics. The mathematical definition may be different for different distribution functions, but the most commonly used are those for Gaussian distributions.

3 Pythagorean means

Aritmetic

$$\text{unweighted:} \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

$$\text{weighted:} \quad \bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

$$\tag{3}$$

Use: computing effective density

Geometric

$$\text{unweighted:} \quad \bar{x} = \left(\prod_{i=1}^{n} x_i\right)^{1/n} \tag{4}$$

$$\text{weighted:} \quad \bar{x} = \left(\prod_{i=1}^{n} x_i^{w_i}\right)^{1/\sum_{i=1}^{n} w_i} \tag{5}$$

$$\tag{6}$$

Use: computing effective thermal conductivity for random oriented mixtures

The arithmetic and geometric means are mathematically related. Suppose we have a set of numbers, $x = (x_1, x_2, \ldots, x_n)$. We can define a tranform of $x \mapsto y$ by $y = \log x$. Arithmetic means are often appropriate when the values of something are often about the same order of magintude, but geometric means make a bit more sense when the range of a property may span orders of magnitude as the do for many types of conductivity (e.g., electrical and hydraulic conductivity).

Harmonic

$$\text{unweighted:} \quad \bar{x} = \left(\sum_{i=1}^{n} x_i^{-1}\right)^{-1} \tag{7}$$

$$\text{weighted:} \quad \bar{x} = \left(\sum_{i=1}^{n} w_i\right)\left(\sum_{i=1}^{n} w_i x_i^{-1}\right)^{-1} \tag{8}$$

Use: computing effective thermal conductivity for layered conductivity packages

4 Other common means

Quadratic mean (RMS)

5 Additional mixing formulae

5.1 Voigt-Reuss-Hill average

The Voigt-Reuss-Hill average is really an average of the volume weighted arithmetic and harmonic means and is common for estimating the bulk or effective magnitude of a physical property. The average is given by

$$\bar{\Psi} = \frac{1}{2}(\Psi_V + \Psi_R) \tag{9}$$

$$= \frac{1}{2}\left[\sum_{i=1}^{n} \Psi_i v_i + \left(\sum_{i=1}^{n} \frac{v_i}{\Psi_i}\right)^{-1}\right], \tag{10}$$

where n is the number of consituent minerals and v_i is the volume fraction of the i-th component.

5.2 Hashin-Strikman bounds

The Hashin-Shtrikman bounds are commonly quoted as a mixing relation for electrical conductivity and occasionally seismic velocity; however, the bounds often over- or under-predict the conductivity, but that is what they are designed to do. The bounds are

$$\sigma_{\mathrm{HS}}^{-} = \mathscr{S}(\sigma_{\min}) \leq \sigma \leq \mathscr{S}(\sigma_{\max}) = \sigma_{\mathrm{HS}}^{+}, \tag{11}$$

where $\mathscr{S}$ is defined as

$$\mathscr{S}(x) = \left(\sum_{i=1}^{N} \frac{\phi_i}{\sigma_i + 2x}\right)^{-1} - 2x, \tag{12}$$

and ϕ_i is the volume fraction of the i-th component.